



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

"A Place to Learn, A Chance to Grow"

Session: 2025-26



VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

UNIT-05: MINING OBJECT, SPATIAL, MULTIMEDIA, TEXT AND WEB DATA

Spatial Data Mining:

- Spatial data mining is a specialized subfield of data mining that deals with extracting knowledge from spatial data.
- Spatial data refers to data that is associated with a particular location or geography.
- Spatial data mining involves analysing and discovering patterns, relationships, and trends in this data to gain insights and make informed decisions.
- Examples of spatial data include maps, satellite images, GPS data, and other geospatial information.
- The use of spatial data mining has become increasingly important in various fields, such as logistics, environmental science, urban planning, transportation, and public health.
- By analysing spatial data, researchers and data mining professionals can identify correlations, predict future events, and make informed decisions that can have a significant impact.

Types of Spatial Data:

Different types of spatial data are used in spatial data mining. These include point data, line data, and polygon data.

❖ **Point Data**

- Point data represents a single location or a set of locations on a map.
- Each point is defined by its x and y coordinates, representing its position in the geographic space.
- Point data is commonly used to represent geographic features such as cities, landmarks, or specific locations of interest.
- Examples of point data in transportation include delivery locations, bus stops, or railway stations.

❖ **Line Data**

- Line data represents a linear feature, such as a road, a river, or a pipeline, on a map.
- Each line is defined by a set of vertices, which represent the start and end points of the line.
- Line data is commonly used to represent transportation networks, such as roads, highways, or railways.

- Line data is also used in other areas, such as hydrology, geology, or ecology, to represent streams, faults, or animal migration routes.

❖ Polygon Data

- Polygon data represents a closed shape or an area on a map.
- Each polygon is defined by a set of vertices that connect to form a closed boundary.
- Polygon data is commonly used to represent administrative boundaries, land use, or demographic data.
- In transportation, polygon data can be used to represent areas of interest, such as delivery zones or traffic zones.

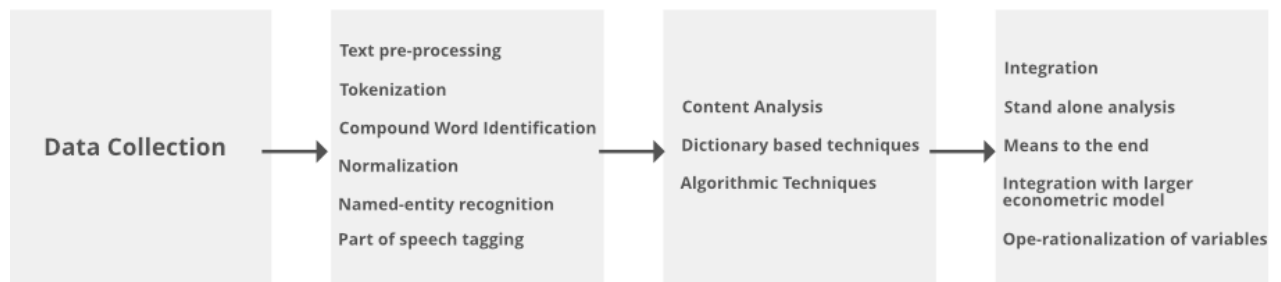
Multimedia Data Mining:

- Multimedia data mining discovers interesting patterns from multimedia databases that store and manage large collections of multimedia objects.
- The Multimedia data includes the image data, video data, audio data, sequence data, hyper-text data containing text.
- Mining multimedia data requires two or more data types, such as text and video or text video and audio.
- Multimedia data mining has a number of uses in today's society an example of this would be the use of traffic camera footage to analyse traffic flow.
- Multimedia data mining can be defined as a process that finds patterns in various types of data, including images, audio, video, and animation.
- Multimedia data mining is an interdisciplinary field that integrates image processing and understanding, computer vision, data mining, and pattern recognition.
- Multimedia data mining discovers interesting patterns from multimedia databases that store and manage large collections of multimedia objects, including image data, video data, audio data, sequence data and hypertext data containing text, text markups, and linkages.
- Issues in multimedia data mining include content-based retrieval and similarity search, generalization and multidimensional analysis.

Text Mining:

- Text mining in data mining is mostly used for, the unstructured text data that can be transformed into structured data that can be used for data mining tasks such as classification, clustering, and association rule mining.
- This allows organizations to gain insights from a wide range of data sources, such as customer feedback, social media posts, and news articles.
- It involves the use of natural language processing (NLP) techniques to extract useful information and insights from large amounts of unstructured text data.
- Text mining can be used as a preprocessing step for data mining or as a standalone process for specific tasks.
- Text mining is widely used in various fields, such as natural language processing, information retrieval, and social media analysis. It has become an essential tool for organizations to extract insights from unstructured text data and make data-driven decisions.

Text Mining Process



Conventional Process of Text Mining

- Gathering unstructured information from various sources accessible in various document organizations, for example, plain text, web pages, PDF records, etc.
- Pre-processing and data cleansing tasks are performed to distinguish and eliminate inconsistency in the data. The data cleansing process makes sure to capture the genuine text, and it is performed to eliminate stop words stemming (the process of identifying the root of a certain word and indexing the data).
- Processing and controlling tasks are applied to review and further clean the data set.
- Pattern analysis is implemented in Management Information System.
- Information processed in the above steps is utilized to extract important and applicable data for a powerful and convenient decision-making process and trend analysis.

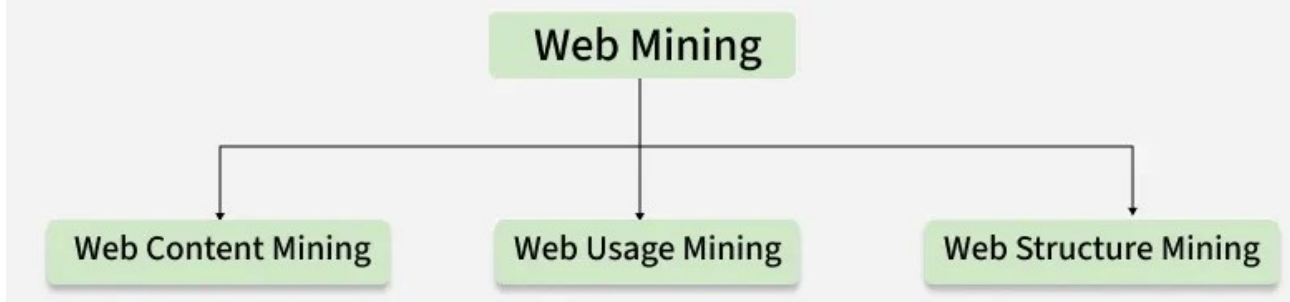
Mining the World Wide Web:

- Web mining is the application of data mining techniques to discover patterns, knowledge, and insights directly from World Wide Web resources, including content, hyperlink structures, and server logs.
- Web mining is the process of applying data-mining, machine-learning and analytical techniques to extract meaningful patterns and insights from the vast data available on the World Wide Web. It aims to discover useful knowledge from web content, structure and user interactions. Its core objective is to convert raw, unstructured web data into actionable information.
- Handles diverse data types including text, images, multimedia, hyperlinks and server logs.
- Combines concepts from data mining, NLP, information retrieval and AI.
- Helps understand user behaviour, website performance and underlying patterns within web ecosystems.
- Works with unstructured, semi-structured and massive, rapidly updating online data.

Types of Web mining

Web mining is broadly classified into three categories based on the type of data being analyzed and the techniques used for analysis.

Types of Web Mining



1. Web Content Mining:

- Web Content Mining focuses on extracting useful information from the actual contents of web pages, including text, images, audio, video and metadata.
- It deals with unstructured or semi-structured data and transforms it into structured forms for analysis.
 - Uses NLP, text mining, multimedia analysis, classification and clustering.
 - Identifies keywords, topics, themes and patterns in documents and media.
 - Helps improve search relevance, content organization and information retrieval.

2. Web Structure Mining

- Web Structure Mining analyzes the link structure of the web to identify relationships between pages and understand how information is connected.
- It treats the web as a directed graph where pages are nodes and hyperlinks are edges
 - Helps identify authoritative or influential pages (e.g., PageRank).
 - Reveals communities, clusters and navigation paths within sites.
 - Useful for SEO, ranking, website design and detecting related content groups.

3. Web Usage Mining

Web Usage Mining deals with analyzing user behaviour by mining web server logs, clickstreams, cookies and session data. It discovers how users navigate, what they prefer and what patterns emerge from usage activity.

- Uses log preprocessing, session reconstruction, pattern mining, clustering and association rules.
- Enables personalization, recommendations, adaptive websites and fraud detection.
- Helps businesses study user journeys, optimize conversions and improve UX.